ROBOTICS & AI ENGINEERING

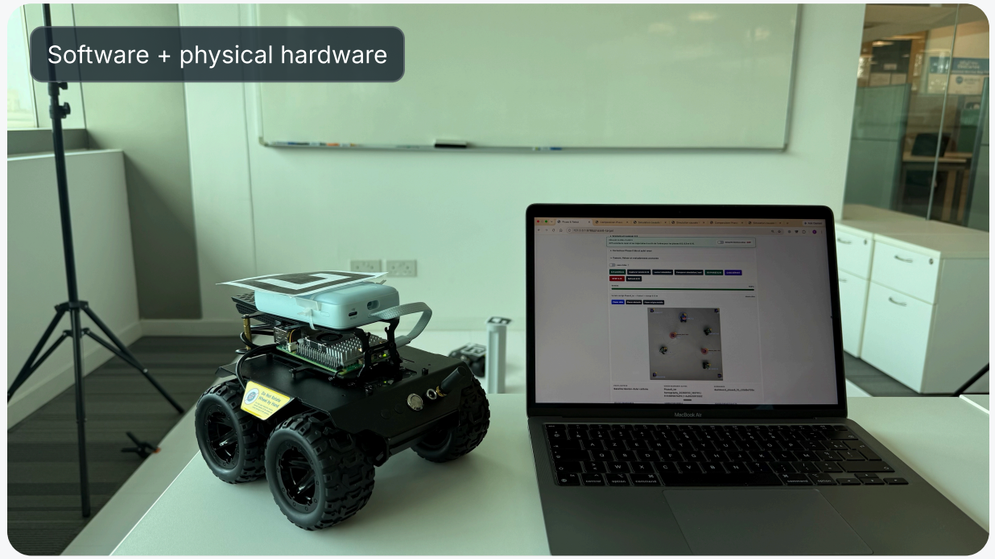
From ideas to working robots.

Slimane Aouanouk

Arts et Métiers

Sorbonne Université

Robotics · Perception · Experiments



Collision avoidance · Singapore

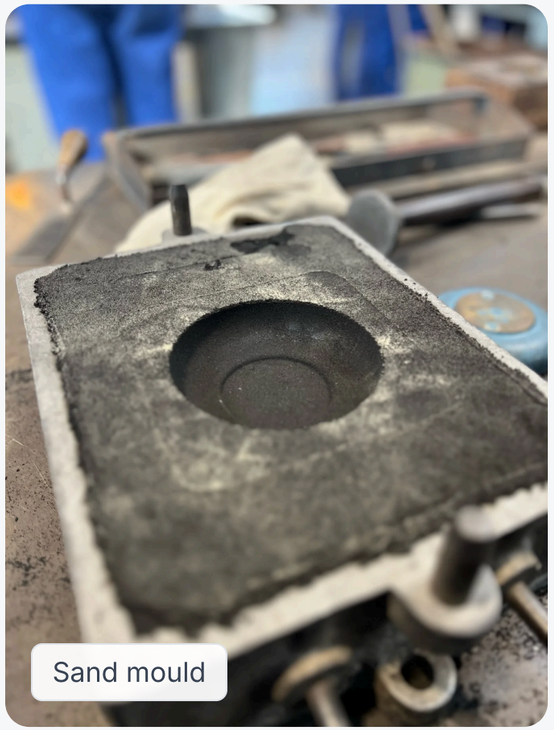
Research internship

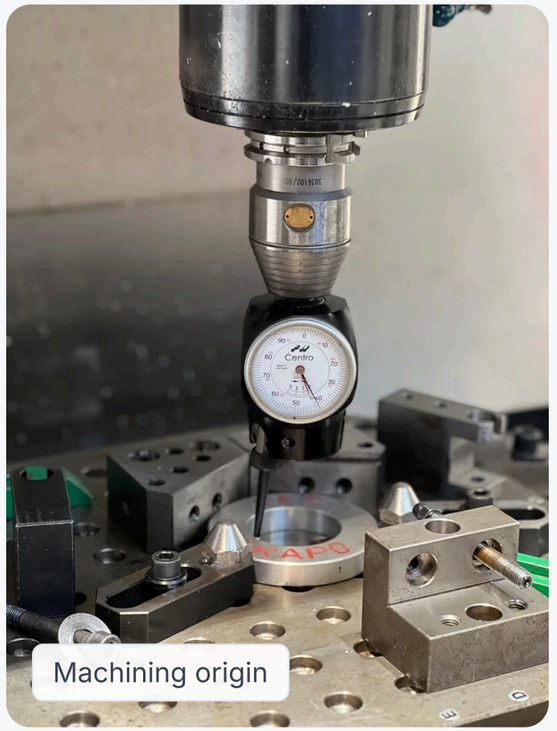
Unitree G1

Ongoing VLA project

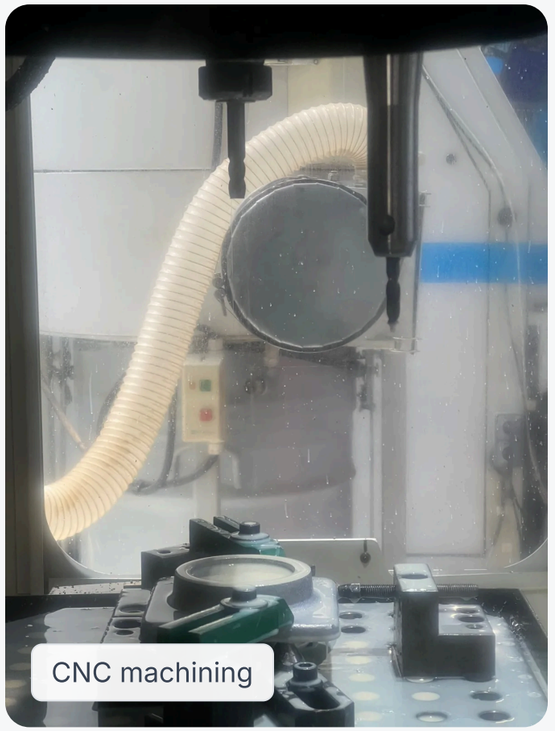
Start with the physical task

Foundry workshop


Sand mould



Machining origin


CNC machining

Foundry → CNC → Inspection • Hands-on manufacturing

• Simulation

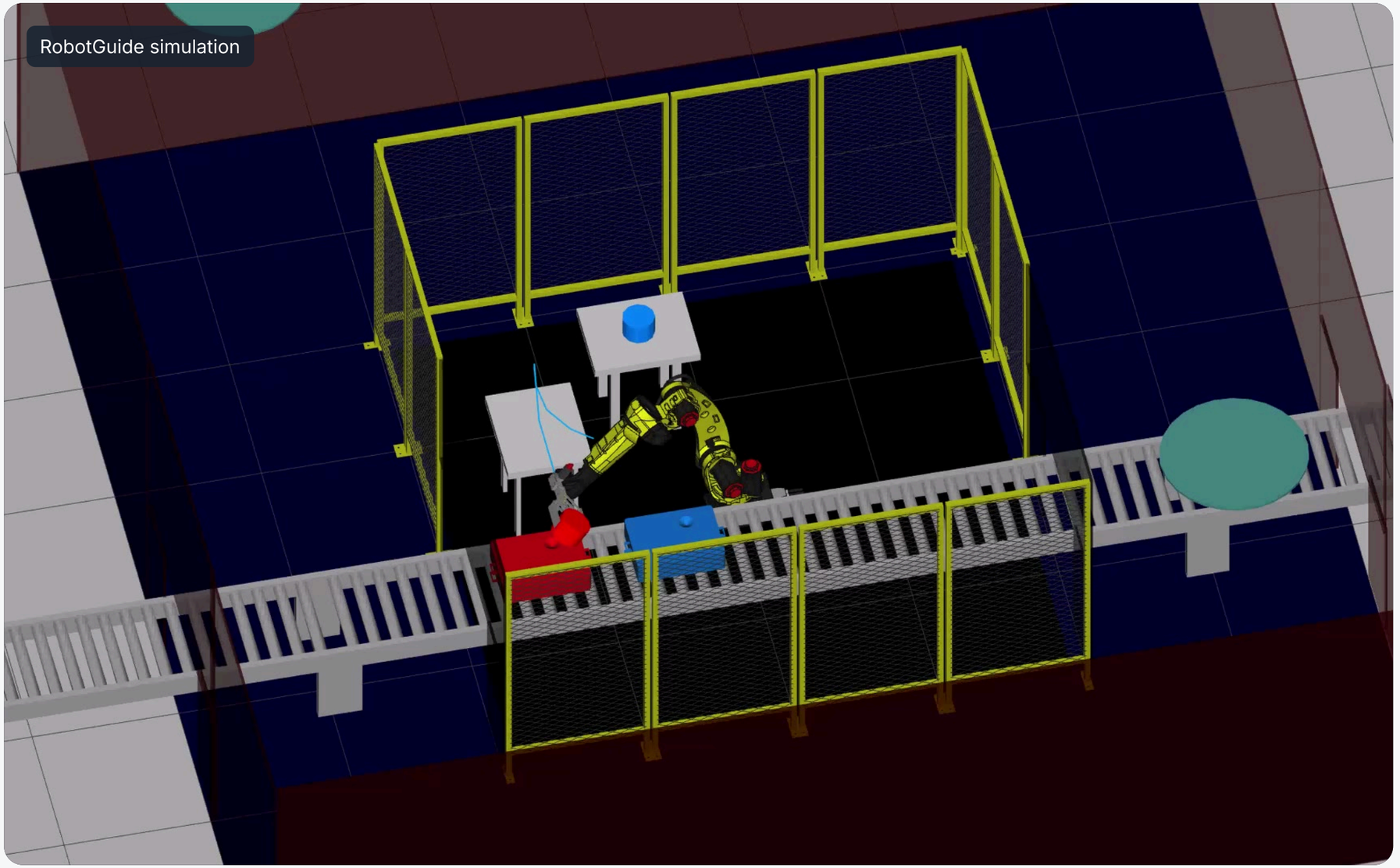
Design the automation

Robot

Gripper

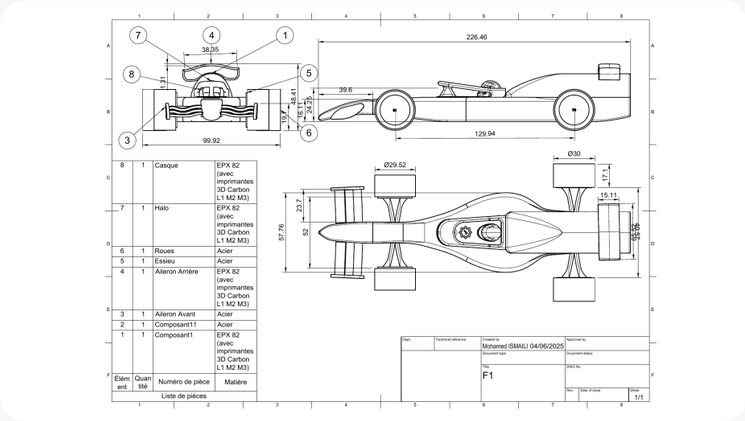
Layout

Sequence



From mechanics to robotics

01 CAD drawing

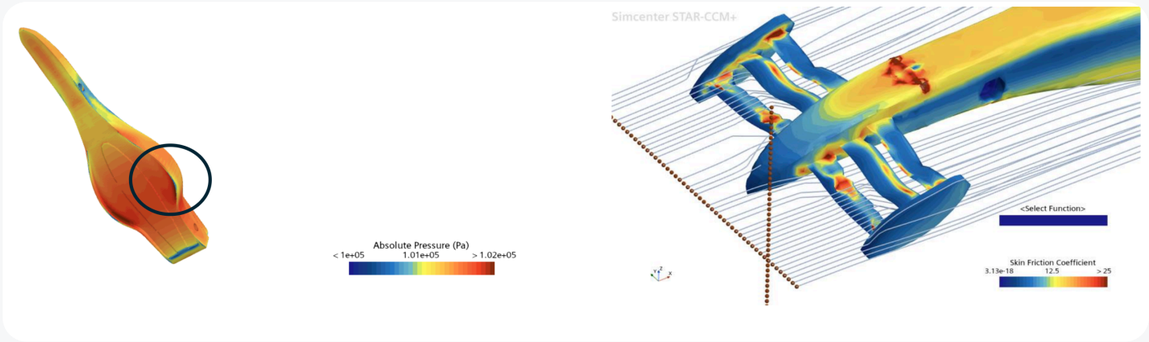


Élément	Quantité	Numéro de pièce	Matière
8	1	Casque	EPX 82 (avec imprimantes 3D Carbon L1 M2 M3)
7	1	Halo	EPX 82 (avec imprimantes 3D Carbon L1 M2 M3)
6	1	Roues	Acier
5	1	Essieu	Acier
4	1	Aileron Arrière	EPX 82 (avec imprimantes 3D Carbon L1 M2 M3)
3	1	Aileron Avant	Acier
2	1	Composant11	Acier
1	1	Composant1	EPX 82 (avec imprimantes 3D Carbon L1 M2 M3)

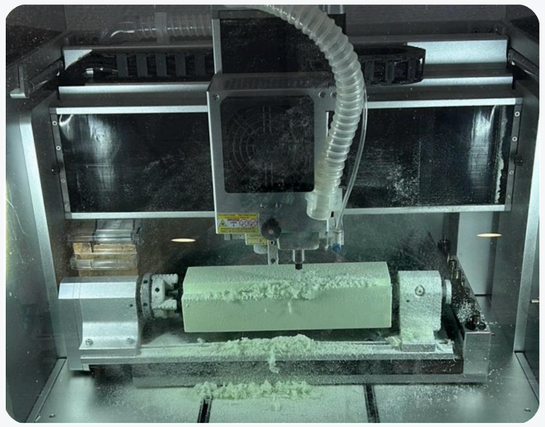
Liste de pièces

→

02 Simulation



03 Foam machining

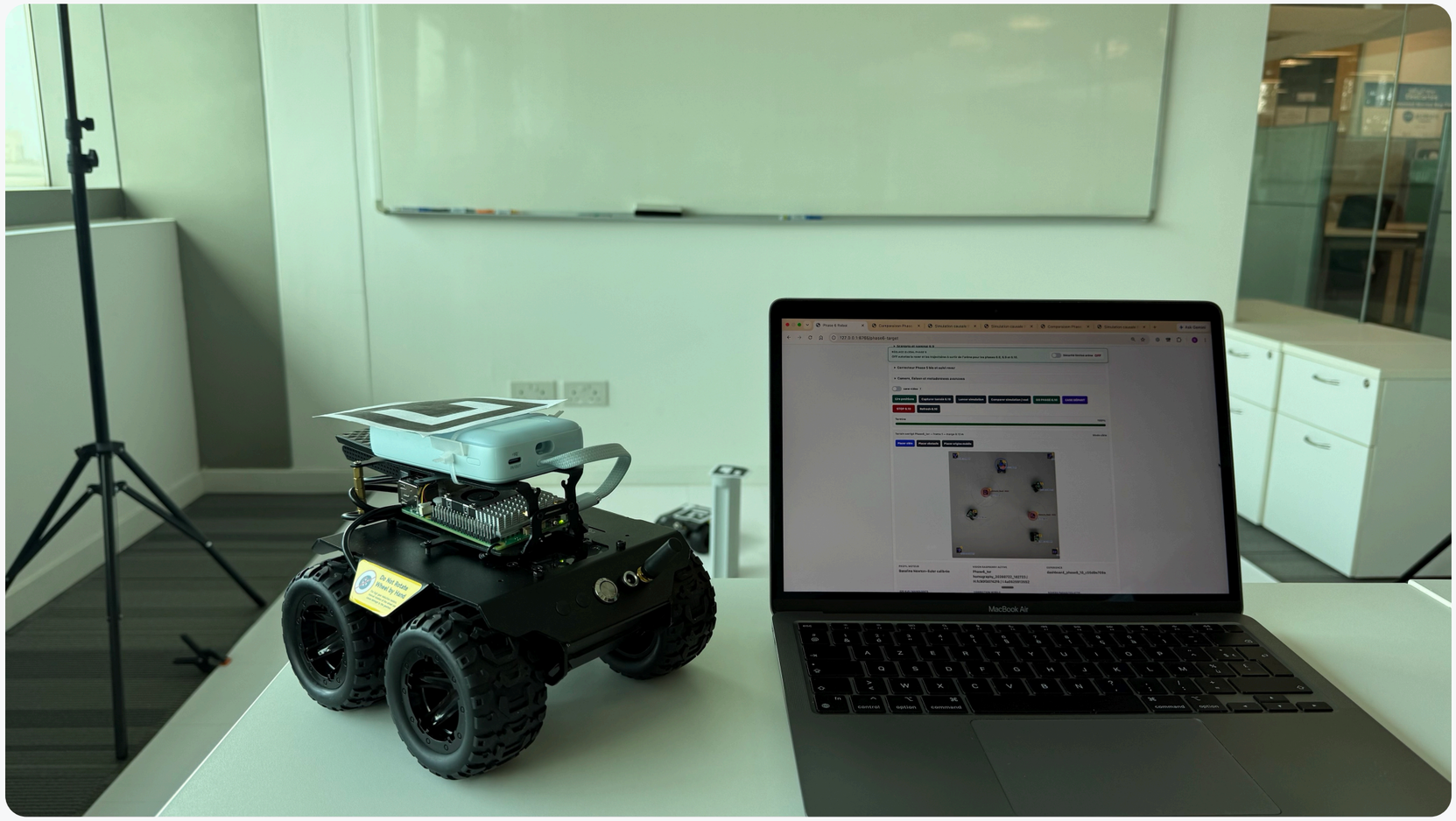


→

04 Finished car

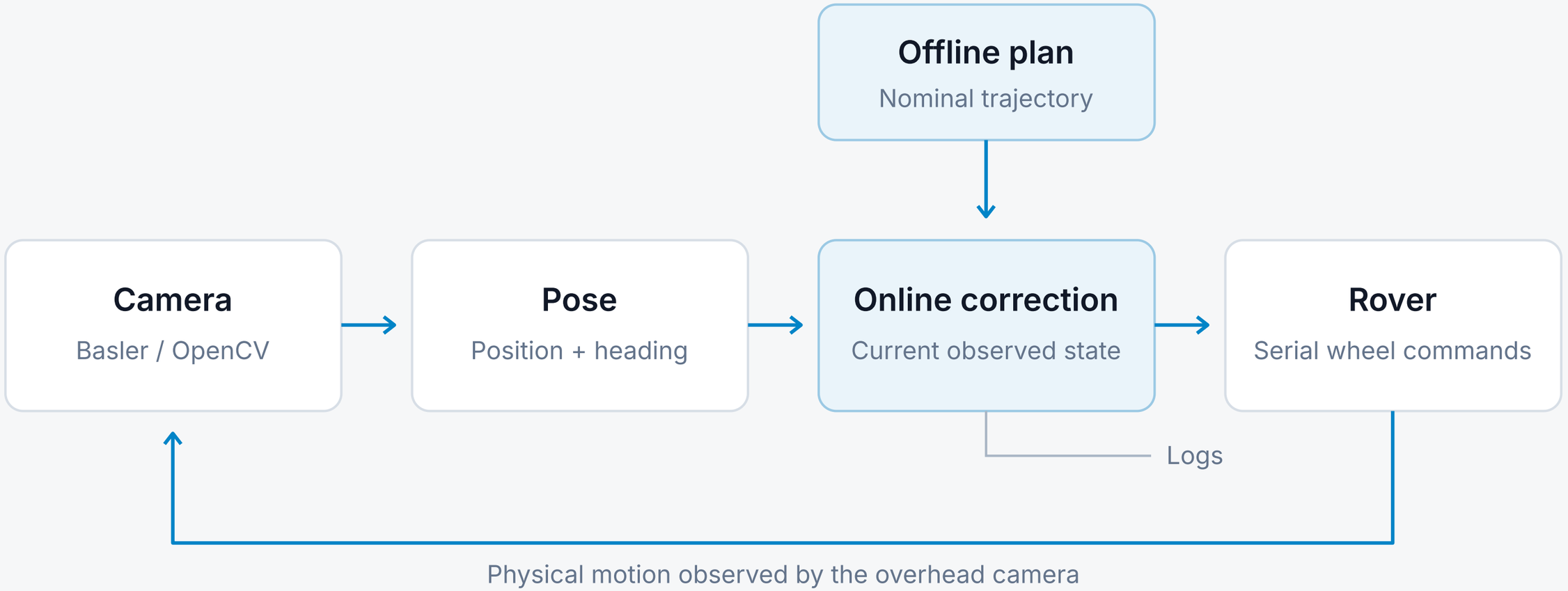
CAD • Simulation • Engineering calculations → Robotics

Make it work on a real robot



Paris → Singapore • Jan–Aug 2026 • Collaborative research

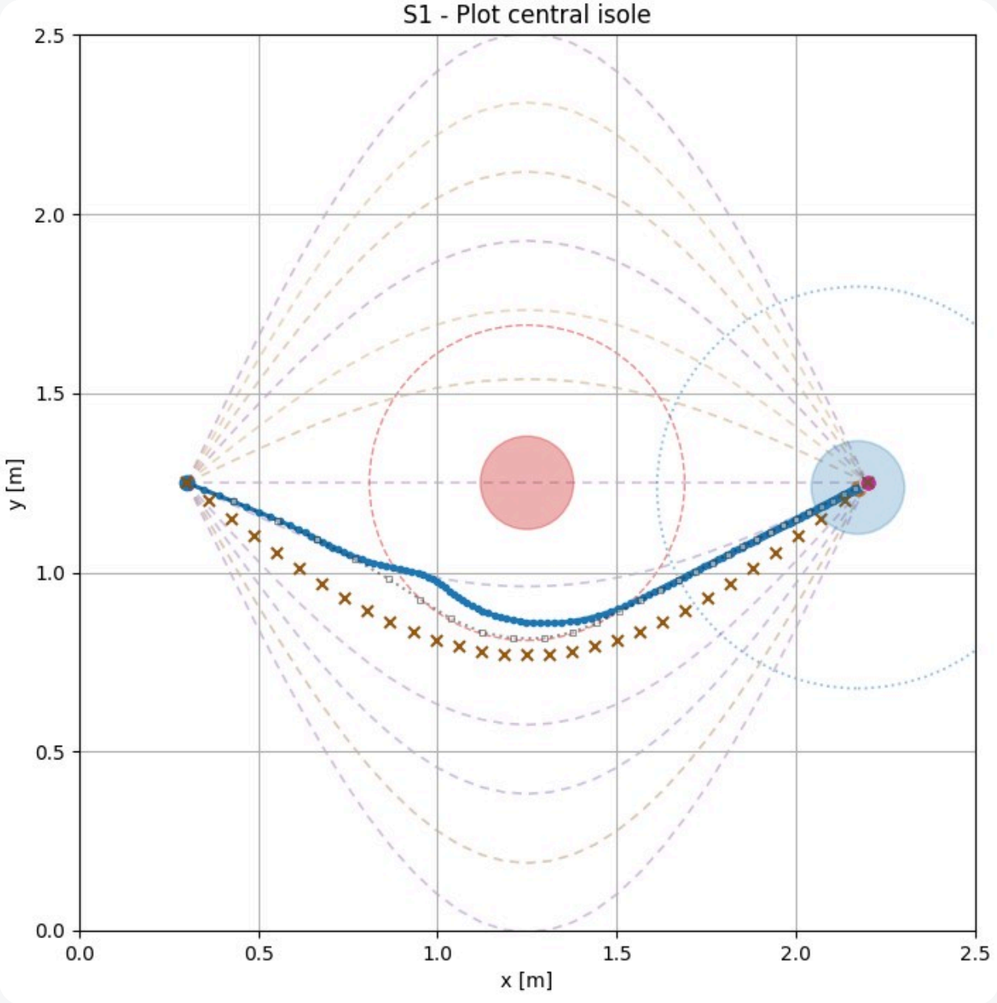
Connect the whole robotics loop



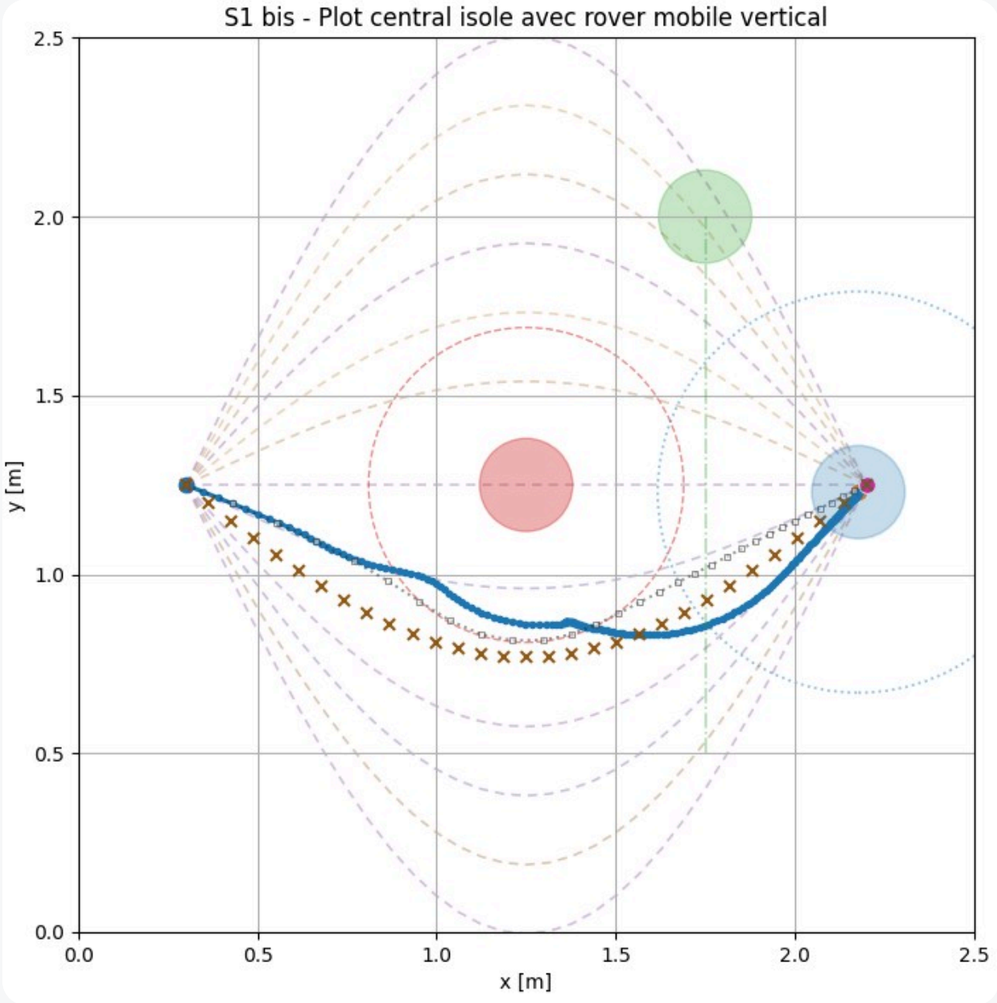
Planning, perception, correction and hardware in one measured system.

Plan the path, then correct the motion

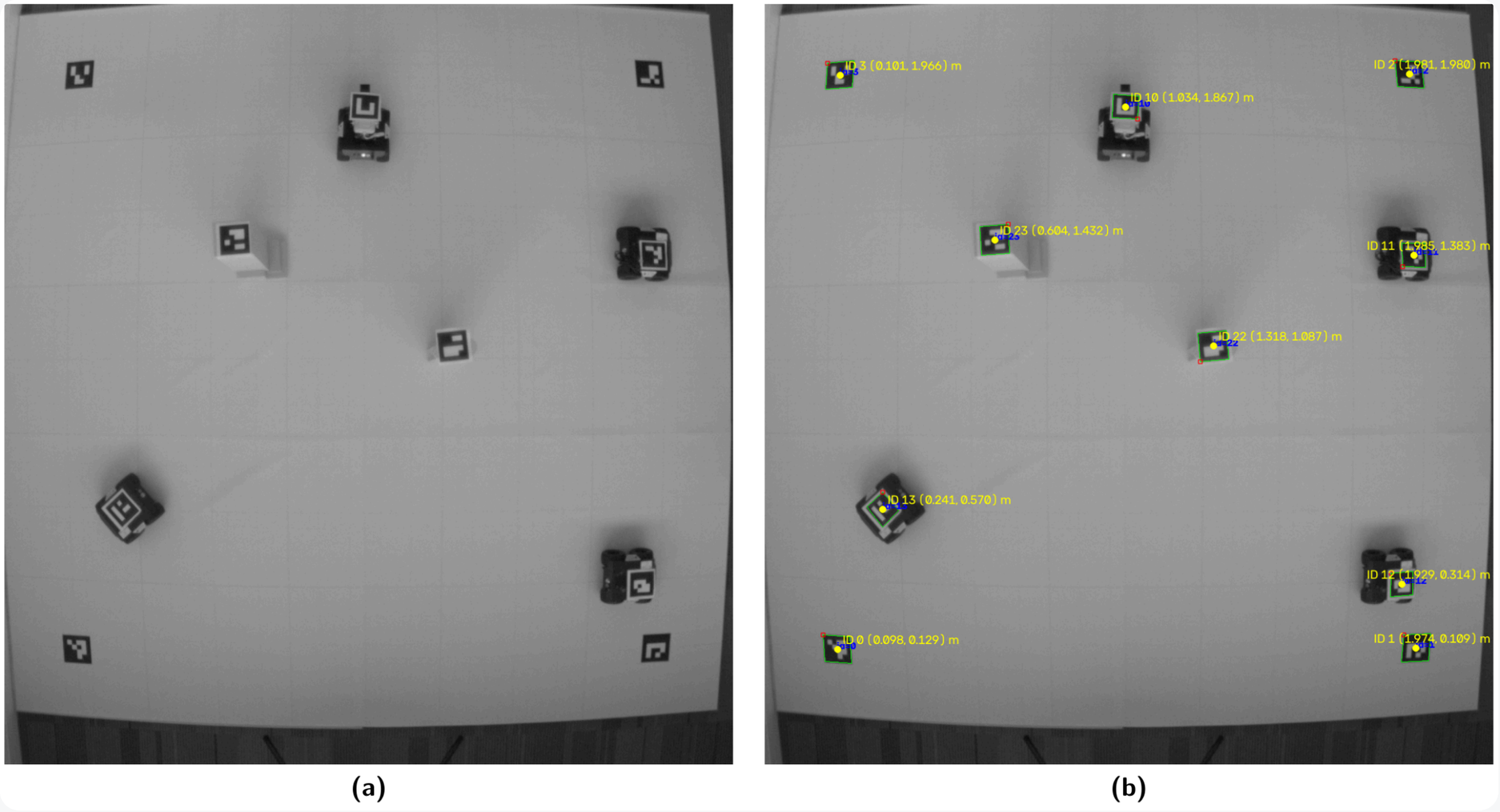
The nominal phase



The causal correction phase



Turn pixels into metric positions



(a) Raw frame • (b) ArUco IDs + metric coordinates • External overhead camera

Improve the measurement, improve the system

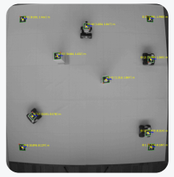
84%

Lower mean
localization error

36.3 mm — Before

5.8 mm — After

0 / Same scale



Independent ground grid
Mean error in millimetres

PROFILE THE COMPLETE SYSTEM

Make sensing and action fit together

≈20 Hz

Measured physical loop

19.84–19.86 Hz median

Vision

Observe the robot

Correction

Use the measured state

Serial commands

Act on physical hardware

REAL EXPERIMENT

Watch the physical system respond

One controlled rover • Moving robots

Check repeatability

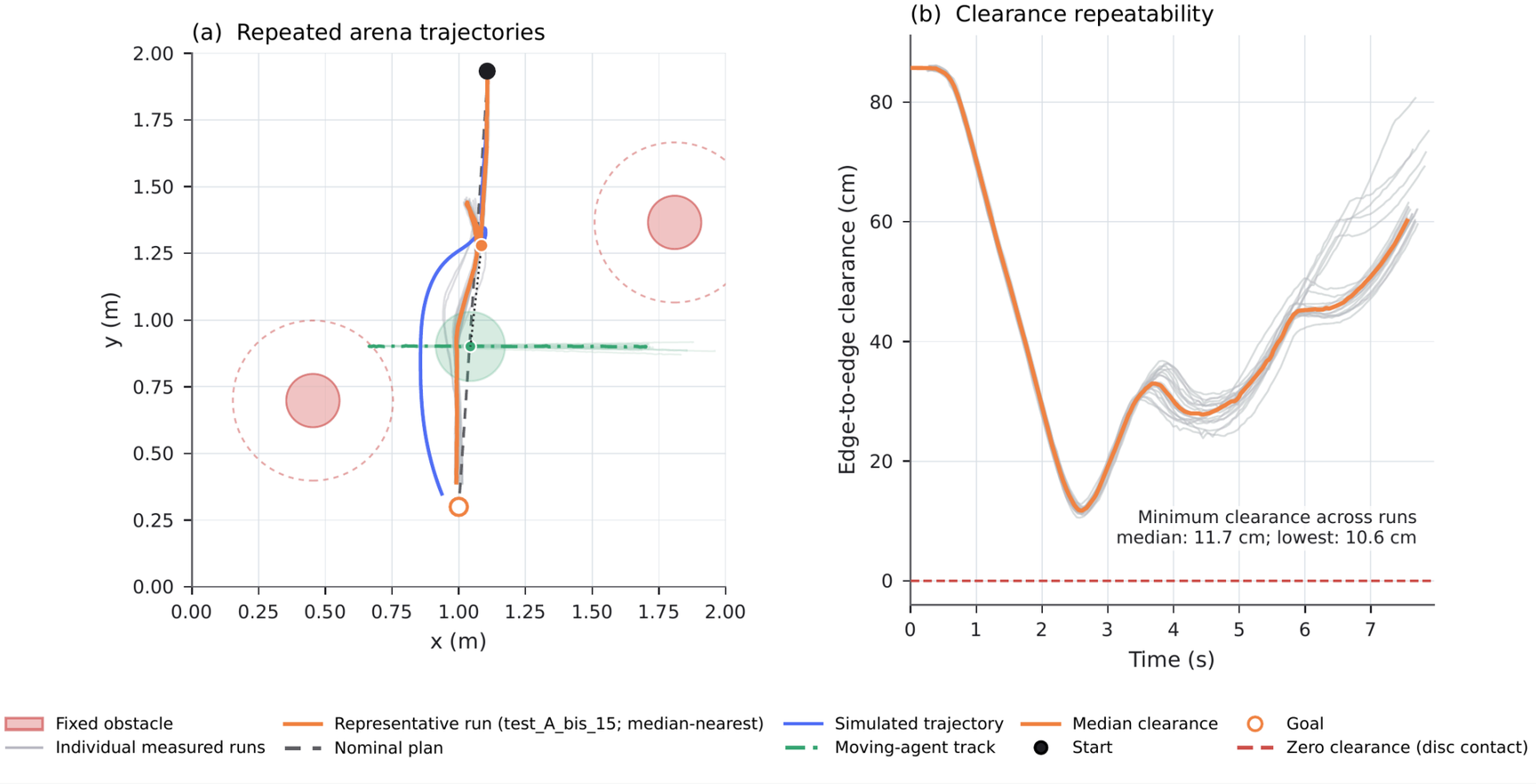
40/40

Goal arrivals

0

Collisions

Two frozen tasks
20 runs each



Shown: one moving agent · 20 physical runs

Measure, diagnose, change, re-test

01

Measure

Grid error

02

Diagnose

Camera geometry

03

Change

Processing

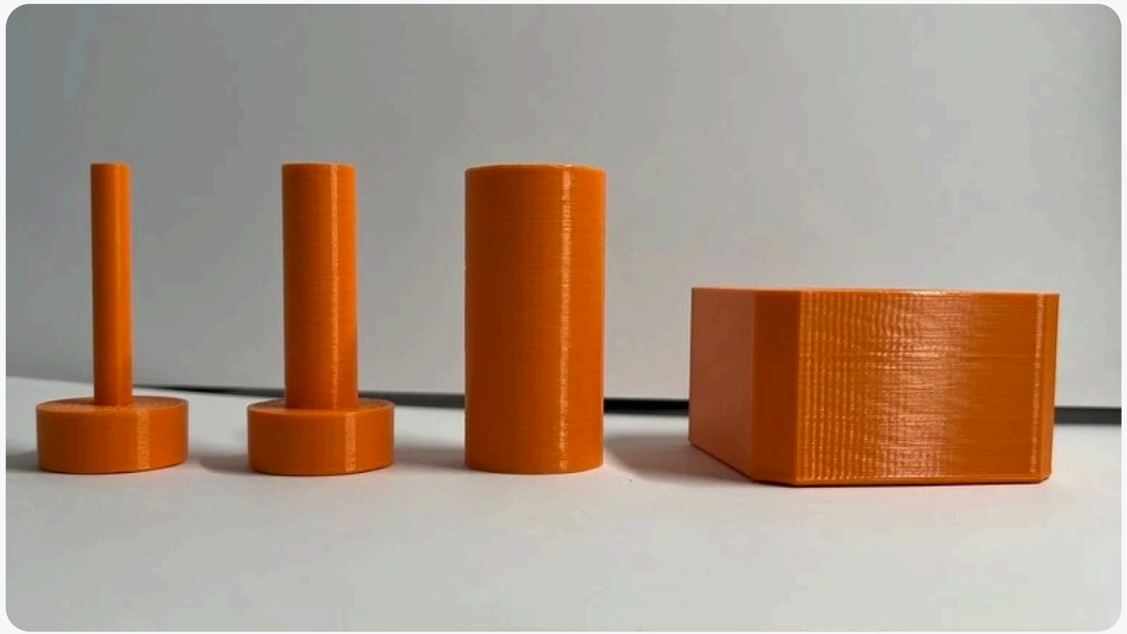
04

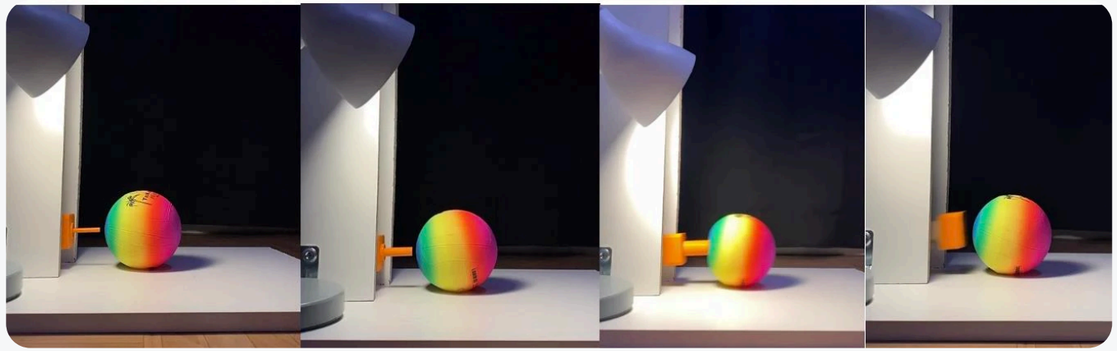
Re-test

Repeatable trials

Geometry · Timing · Software

Build the experiment, analyze the data





Build → Film → Analyze in Python • 46 cm / 4 contact geometries

• Current coursework

What I’m learning now

State estimation & sensor fusion

Perception & mobile robotics

Manipulation & control

Machine learning

Master project → **VLA**

SORBONNE UNIVERSITÉ / MASTER PROJECT

• Ongoing

Bring that mindset to humanoid learning

Unitree G1

Assistive manipulation

Vision + Instruction

Observations and task

→

VLA policy

Learning from demonstrations

→

Robot action

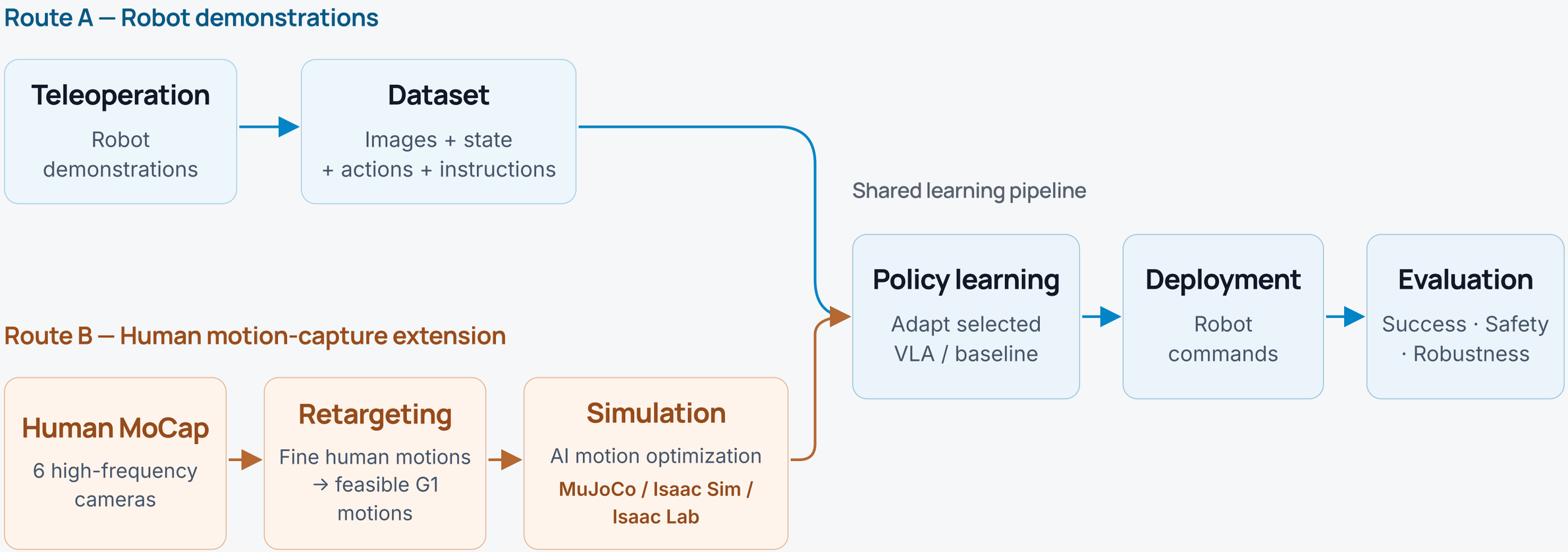
Assistive manipulation

- Literature review → Project setup

Supervised by Prof. Olivier Bruneau

• Project direction

From demonstrations to robot behavior



A POSSIBLE CONNECTION WITH BINABIK

Use the project to prepare for the internship

01 / DISCUSS

Your priorities

02 / EXPLORE

A focused experiment

Demonstration quality

Policy evaluation

03 / PREPARE

A stronger start

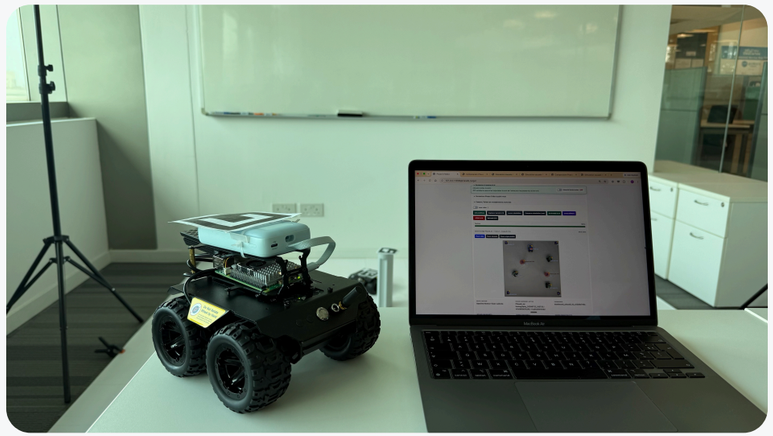
An opportunity to explore a useful direction before joining the team.

WHERE THE EXPERIENCE CONNECTS

Why Binabik

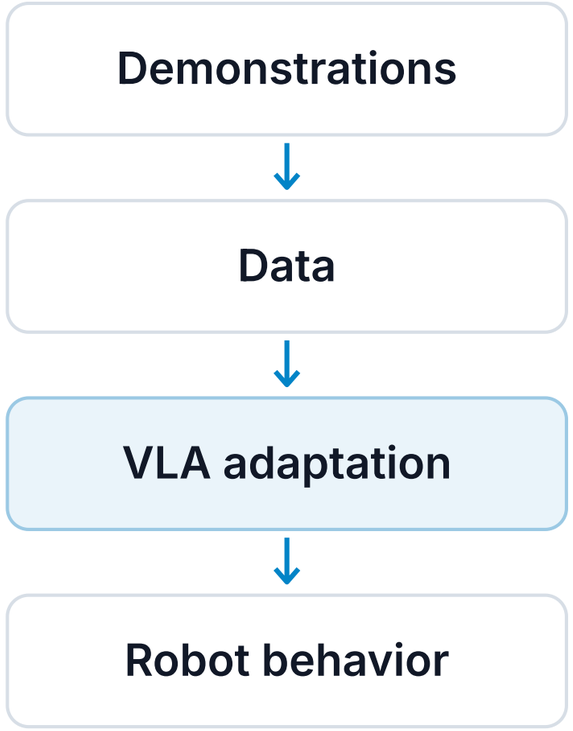
Industrial tasks

Understand the task



Real robots

Integrate and measure



Learning-based behavior

Develop adaptability

LET’S DISCUSS WHERE I COULD BE USEFUL

Where I’d like to contribute

- Perception & integration
- Robot learning & data
- Experiments & debugging

6-month internship / February 2027

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VLA project: current work and directions

Unitree G1 / Assistive manipulation

Current work

Literature review & project setup

Demonstration quality

Policy evaluation

POSSIBLE ACQUISITION ROUTE

Motion capture → Retargeting

Proposed by Prof. Olivier Bruneau.
Future six-camera system.

Possible motion optimization in MuJoCo / Isaac.

- To be explored

Exploring Synthetic Data Generation for Humanoid Robotics

NVIDIA 2026-5-28

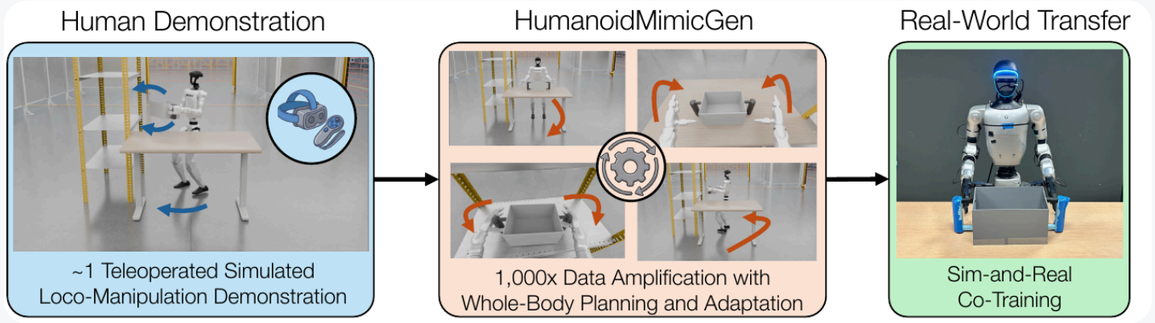
HumanoidMimicGen: Data Generation for Loco-Manipulation via Whole-Body Planning

Kevin Lin[1,2,*], **Ajay Mandlekar**[1,*], **Caelan Reed Garrett**[1,*], **Nikita Chernyadev**[1], **Yu Fang**[1], **Runyu Ding**[1], **Yuqi Xie**[1], **Justin Tran**[1], **Linxi Fan**[1,†], **Yuke Zhu**[1,2,†]

[1]NVIDIA [2]The University of Texas at Austin

[*] Equal contribution [†] Project leads

https://humanoidmimicgen.github.io/



Lin et al. · 2026 · Paper & project

HumanoidMimicGen

NVIDIA & UT Austin · 2026

Challenge

Limited real-world demonstrations

Approach

Synthetic data generation through whole-body planning

Key idea

1 teleoperated demo → 1,000 synthetic demos

Per task in the simulation benchmark

Potential impact

Less data collection effort when adapting to new tasks

My interest How can we reduce the data collection effort when deploying robots to new industrial tasks?

APPENDIX A3

Research interests: from VLA to reliable robots

Policy adaptation

How can we adapt a pretrained policy?

OpenVLA · Ψ_0

Data efficiency

How can we reduce demonstration collection?

HumanoidMimicGen · Ψ_0

Contact precision & reliability

How can we diagnose and reduce failures?

FACT

Whole-body execution

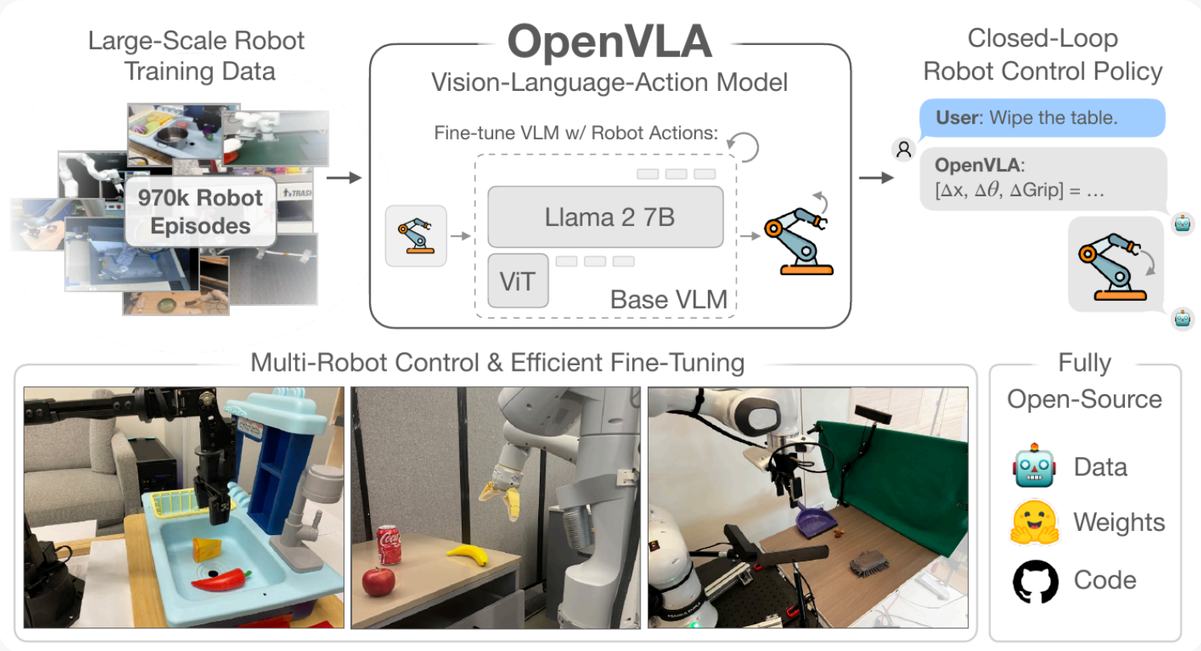
How can high-level decisions become stable movements?

LeVERB

For your industrial tasks

Where is the biggest bottleneck today: data, policy adaptation, or reliable execution?

Research interests: from VLA to reliable robots



Kim et al. · 2024 · Fig. 1, p. 1 · Paper / project

OpenVLA

Research question

How can we adapt a pretrained policy to a new task?

Main idea

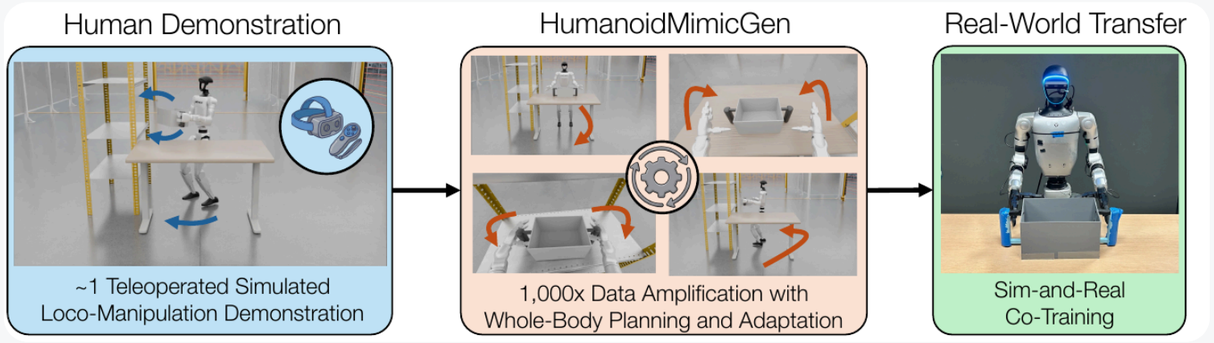
Fine-tune a pretrained VLA using robot demonstrations.

Why it interests me

Understand policy adaptation for my VLA project.

Scope Manipulation policies; not full humanoid locomotion or whole-body control.

Research interests: from VLA to reliable robots



Lin et al. · 2026 · Fig. 1, p. 2 · Paper / project

HumanoidMimicGen

Research question

How can we expand a small set of demonstrations?

Main idea

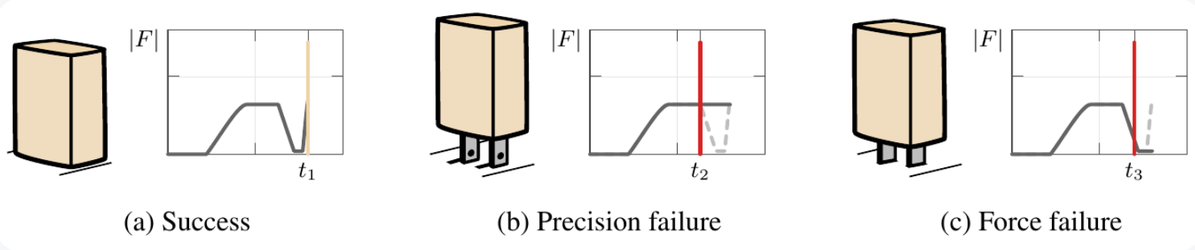
Generate synthetic demonstrations through whole-body planning and adaptation.

Why it interests me

Explore data efficiency when adapting the G1 to new scenes.

Scope 1 source demo → 1,000 generated demos per task in simulation; real-world evaluation also uses real data. More on this example → A2

Research interests: from VLA to reliable robots



(a) Success (b) Precision failure (c) Force failure

Parés-Morlans et al. · 2026 · Fig. 2, p. 4 · Paper / project

FACT

Research question

Why do VLA policies fail during contact?

Main idea

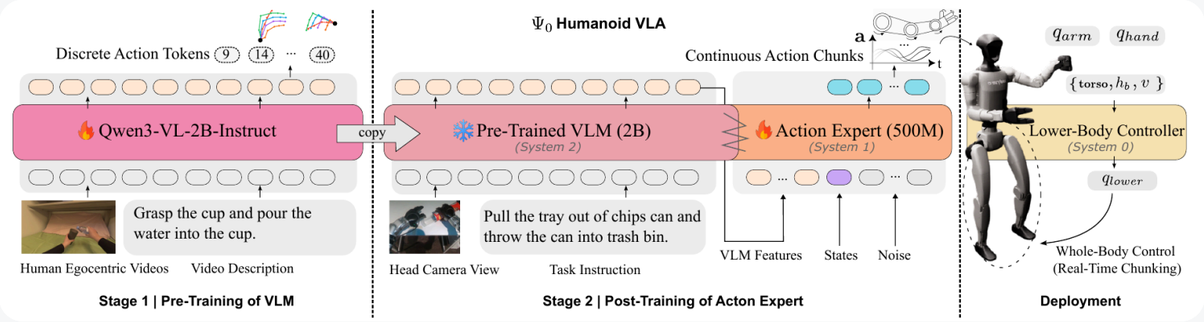
Separate precision failures from force failures; improve fine actions and force conditioning.

Why it interests me

Build reproducible tests and diagnose failure modes.

Scope Contact-rich manipulation with suitable force sensing; not a general reliability guarantee.

Research interests: from VLA to reliable robots



Wei et al. · 2026 · Fig. 2, p. 3 · Paper / project

Ψ_0 — Psi-Zero

Research question

How can human videos support humanoid learning?

Main idea

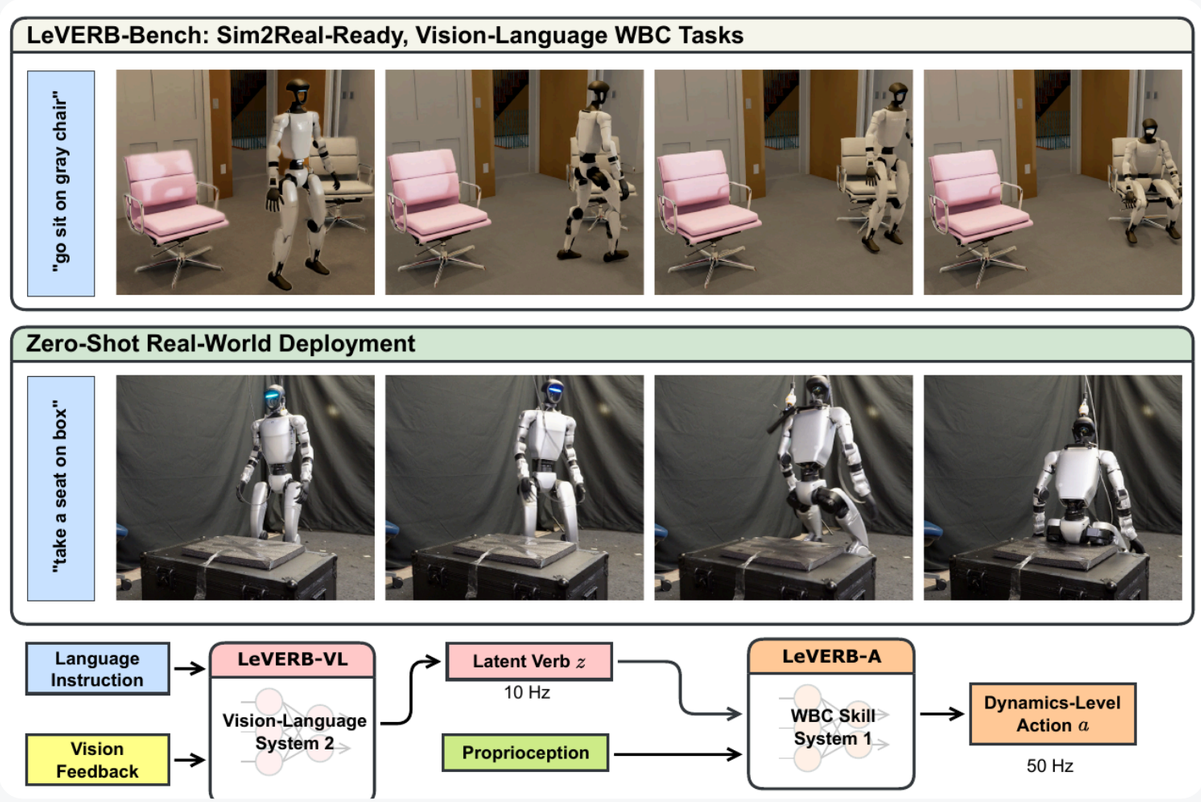
Pretrain on human videos, then learn robot actions from robot data.

Why it interests me

Understand how human demonstrations can inform robot learning.

Scope Human video pretraining is distinct from motion capture and does not directly supply robot joint commands.

Research interests: from VLA to reliable robots



Xue et al. · 2025 · Fig. 1, p. 2 · Paper / project

LeVERB

Research question

How can high-level instructions produce stable movements?

Main idea

Link a vision-language policy to a whole-body controller through latent actions.

Why it interests me

Connect robot learning with physical control and integration.

Scope Whole-body behaviors and control; not a method demonstrated for fine industrial manipulation.

Calibration: which error are we measuring?

Reprojection RMS

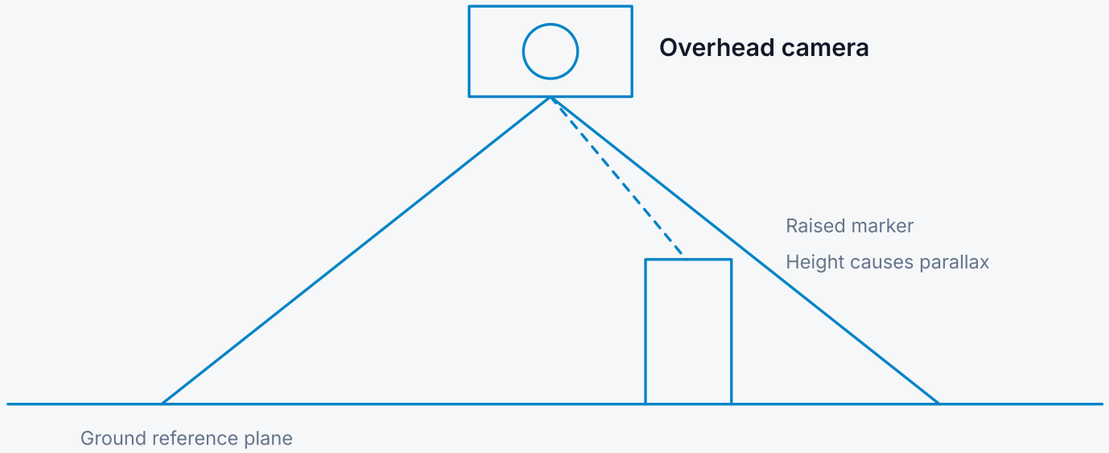
0.191 px

Ground-grid mean error

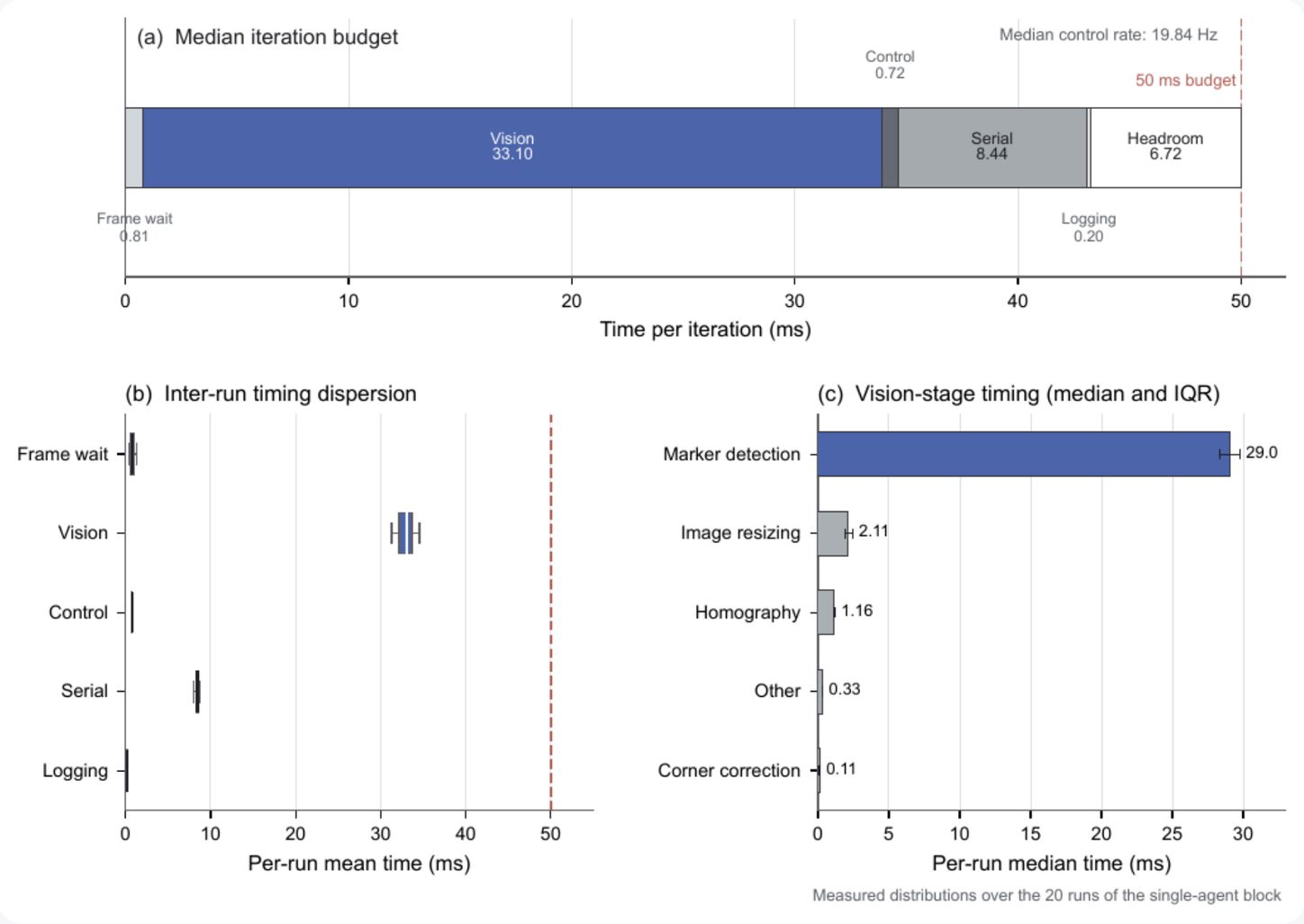
36.3 → 5.8 mm

Raised-marker mean error

60.8 → 11.6 mm



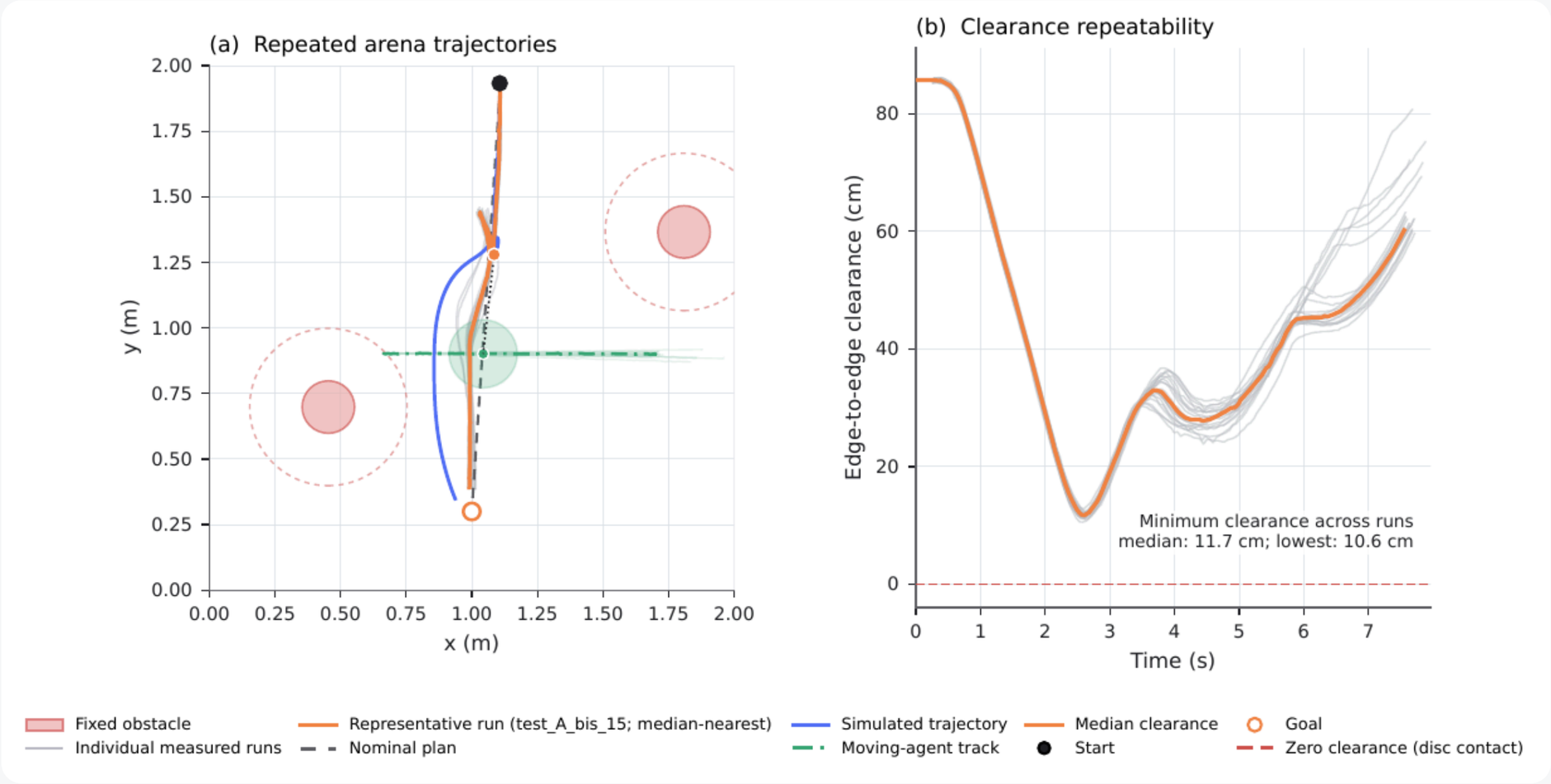
What sits inside a near-20 Hz loop?



Vision · Correction · Serial I/O · Logging • Observed medians

Repeatability and limits

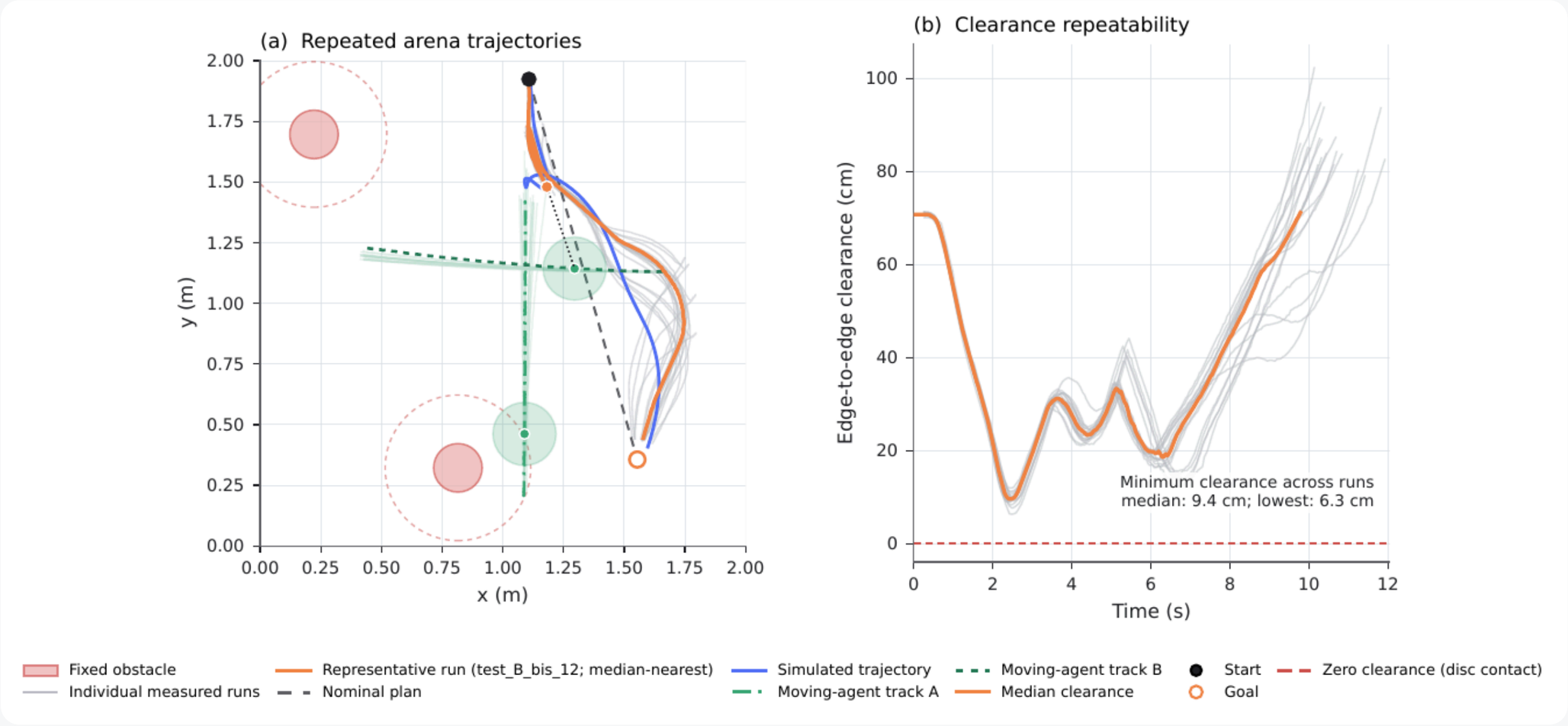
One moving agent | Two moving agents



Frozen task / 20 physical runs. Clearance is defined between safety discs.

Repeatability and limits

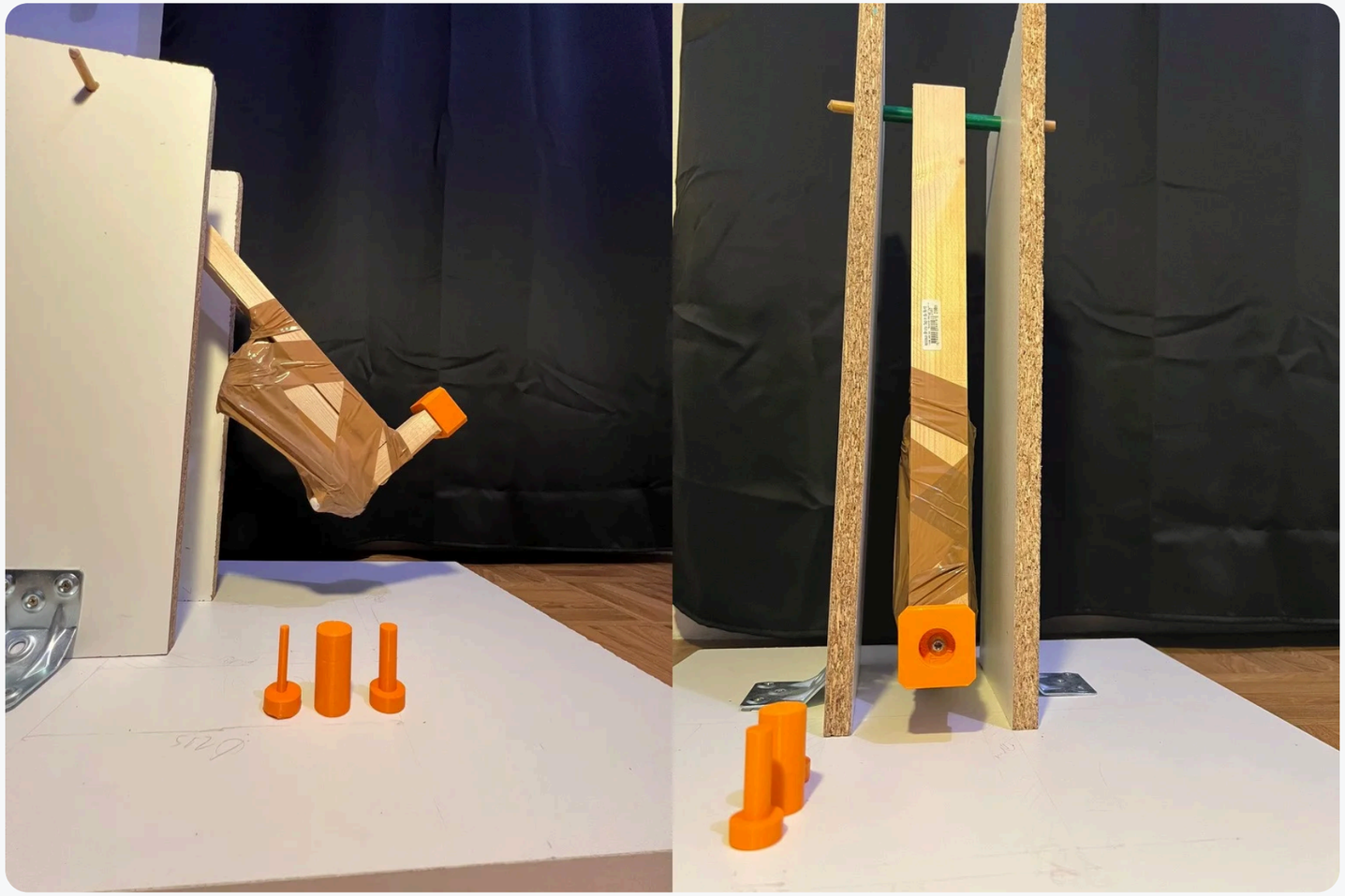
One moving agent | Two moving agents



19 runs in the aggregate after one layout exclusion. The excluded run still reached its goal.

Individual experimental work and physical design

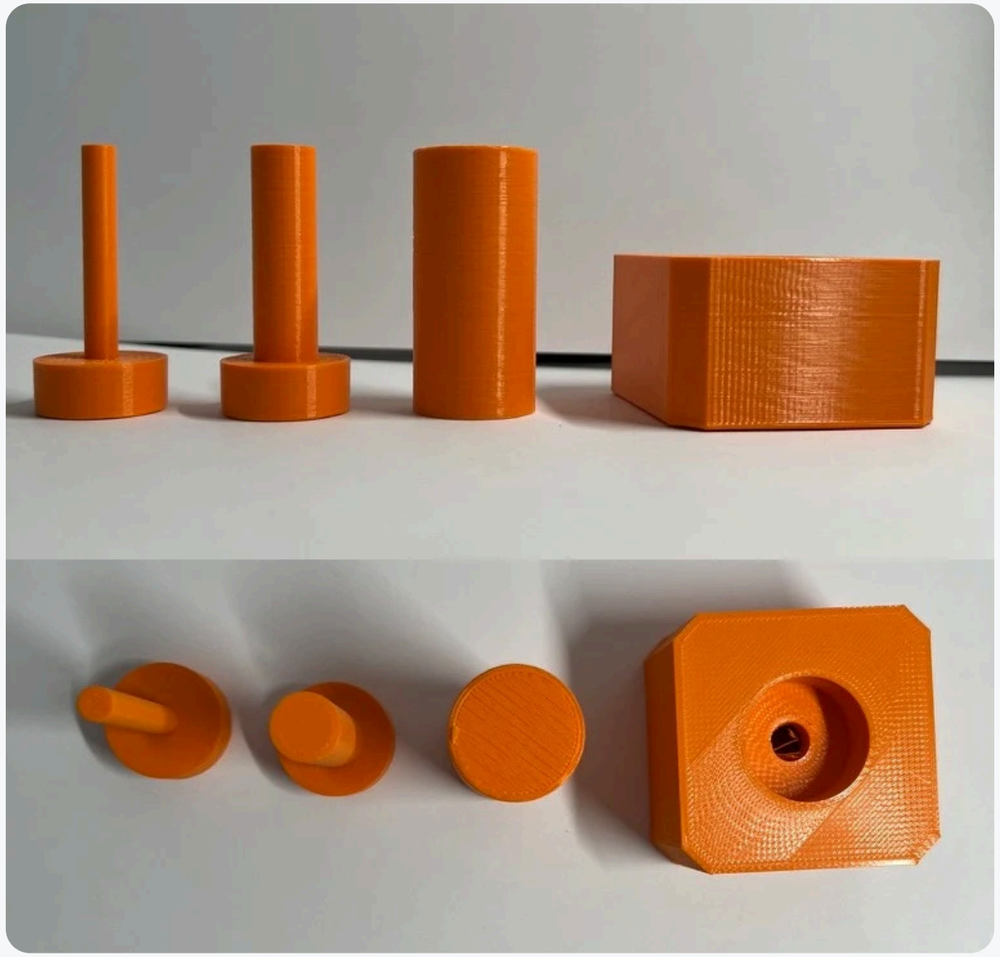
TIPE rig | Contact geometries | Regressions | Speed comparison | CAD design | Manufacturing



Individual TIPE / 46 cm pendulum / Video-based measurements

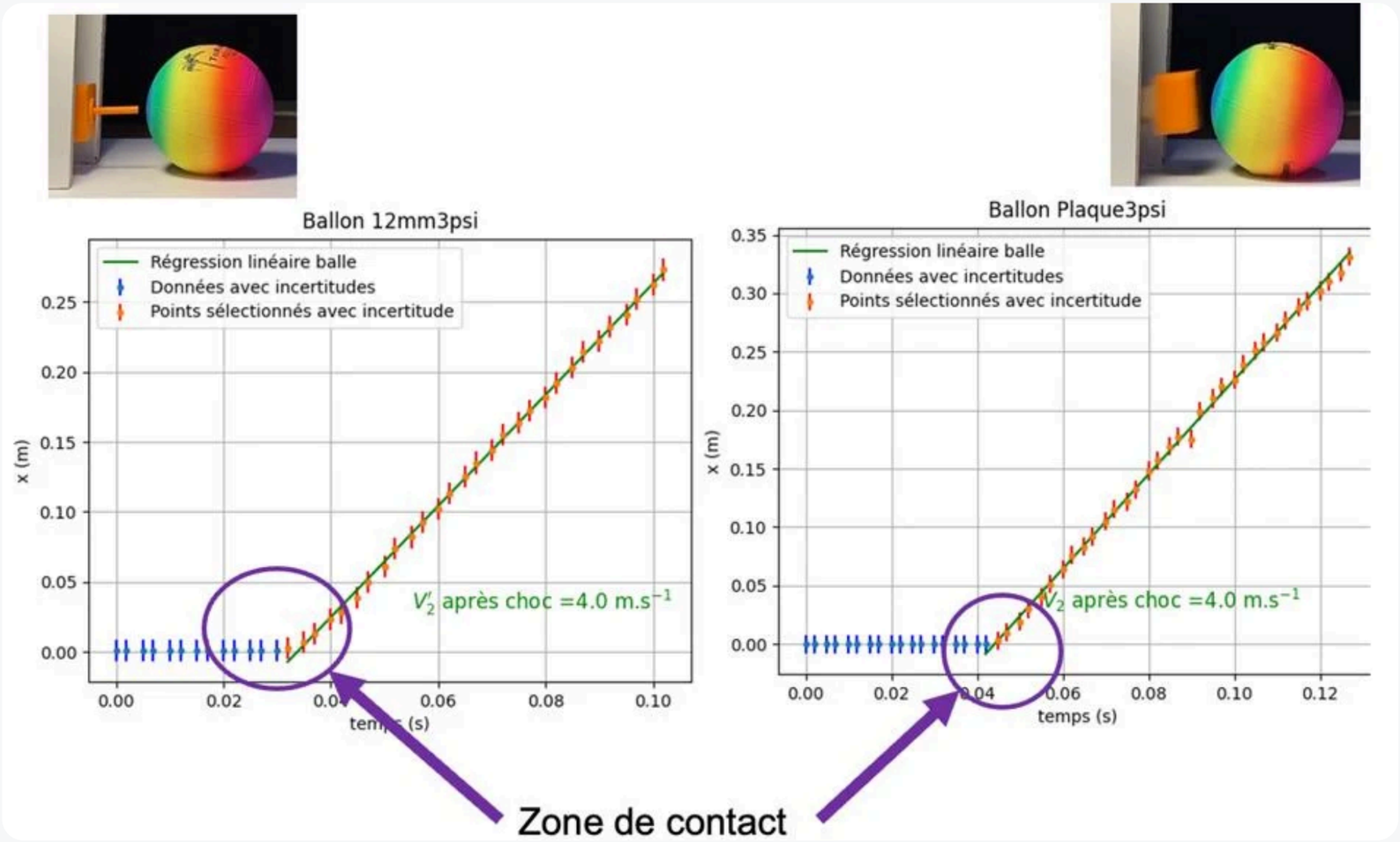
Individual experimental work and physical design

TIPE rig | Contact geometries | Regressions | Speed comparison | CAD design | Manufacturing



Individual TIPE / Four contact geometries

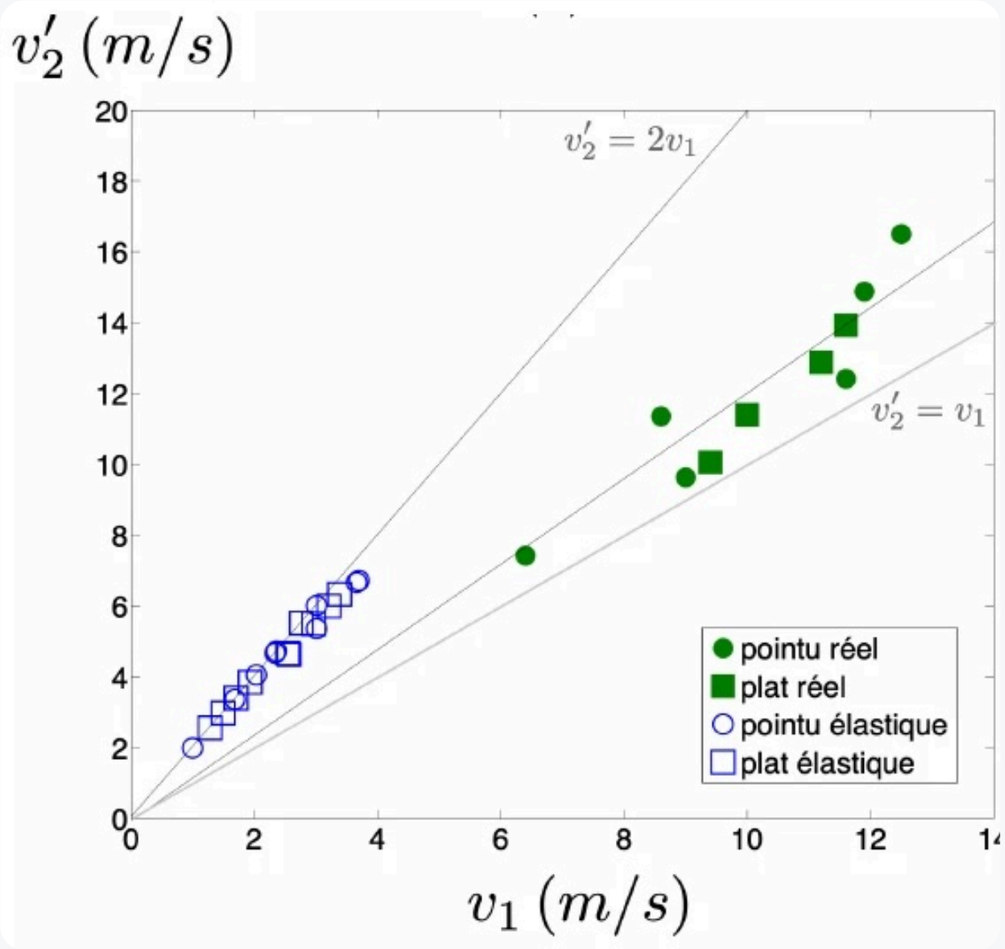
Individual experimental work and physical design



Individual TIPE / Position fits before and after impact, with uncertainty

Individual experimental work and physical design

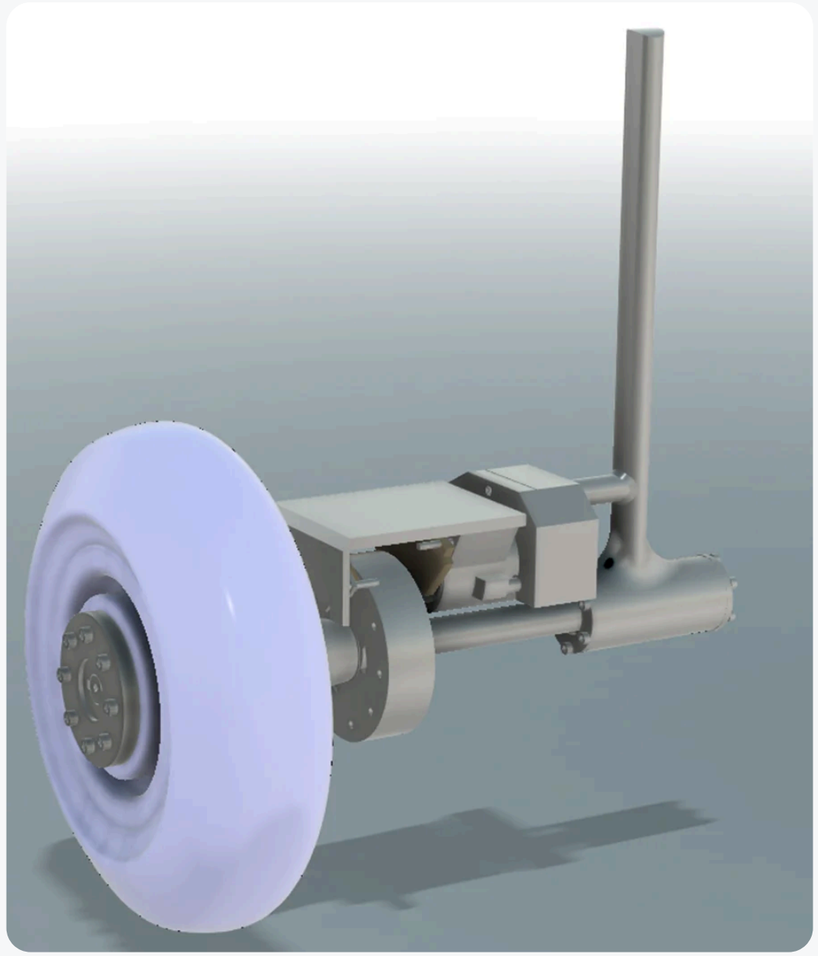
TIPE rig | Contact geometries | Regressions | Speed comparison | CAD design | Manufacturing



Individual TIPE / Interpret the experimental conditions before generalizing

Individual experimental work and physical design

TIPE rig | Contact geometries | Regressions | Speed comparison | CAD design | Manufacturing



Individual preliminary design / Fusion 360 / No validated physical prototype

Individual experimental work and physical design

TIPE rig | Contact geometries | Regressions | Speed comparison | CAD design | Manufacturing

Team manufacturing project / Nonconformities diagnosed and documented